

Q1 - 24 January - Shift 1

Vectors $a\hat{i} + b\hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is _____.

Space for your notes:

Q2 - 24 January - Shift 2

If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other.

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Then, the value of m will be :

- (1) 1 (2) -1
(3) -3 (4) 2

Q3 - 25 January - Shift 1

If $\vec{P} = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $\vec{Q} = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$ then,

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The unit vector in the direction of $\vec{P} \times \vec{Q}$ is $\frac{1}{x}(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k})$. The value of x is _____.

Q4 - 29 January - Shift 2

In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab at 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is _____

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (1)

Q2 (4)

Q3 (4)

Q4 (41)

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#MathBoleTohMathonGo

Q1 (1)

For two perpendicular vectors

$$(a\hat{i} + b\hat{j} + \hat{k}) \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 0$$

$$2a - 3b + 4 = 0$$

On solving, $2a - 3b = -4$

Also given

$$3a + 2b = 7$$

We get $a = 1, b = 2$

$$\frac{a}{b} = \frac{x}{2} \Rightarrow x = \frac{2a}{b} = \frac{2 \times 1}{2}$$

$$\Rightarrow x = 1$$

Q2 (4)

$$\vec{P} \cdot \vec{Q} = 0$$

$$(\hat{i} + 2m\hat{j} + m\hat{k}) \cdot (4\hat{i} - 2\hat{j} + m\hat{k}) = 0$$

$$\Rightarrow 4 - 4m + m^2 = 0$$

$$\Rightarrow (m - 2)^2 = 0 \Rightarrow m = 2$$

Q3 (4)

$$\vec{P} \times \vec{Q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & \sqrt{3} & 2 \\ 4 & \sqrt{3} & 2.5 \end{vmatrix} = \sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k}$$

$$\Rightarrow \frac{\vec{P} \times \vec{Q}}{|\vec{P} \times \vec{Q}|} = \frac{1}{2} \left(\sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k} \right)$$

$$= \frac{1}{4} (\sqrt{3} \hat{i} + \hat{j} - 2\sqrt{3} \hat{k}) \quad x = 4$$

Q4 (41)

$$u = \frac{h}{h'} = \frac{5.25}{5.00}$$

$$\text{Least count} = \frac{1}{20} \text{ cm} - \frac{49}{50} \cdot \frac{1}{20} \text{ cm}$$

$$= \frac{1}{50} \times \frac{1}{20} \text{ cm} = 0.01 \text{ mm}$$

$$\ln u = \ln h - \ln h'$$

$$\frac{du}{u} = \frac{dh}{h} - \frac{dh'}{h'}$$

$$du = \left[\frac{0.01}{5.25} + \frac{0.01}{5.00} \right] \frac{5.25}{5.00}$$

$$= \frac{41}{10} \times 10^{-3}$$

$$\text{Ans.} = 41$$